

Revenue-Aware ECC Routing — Deployment Plan & Architecture

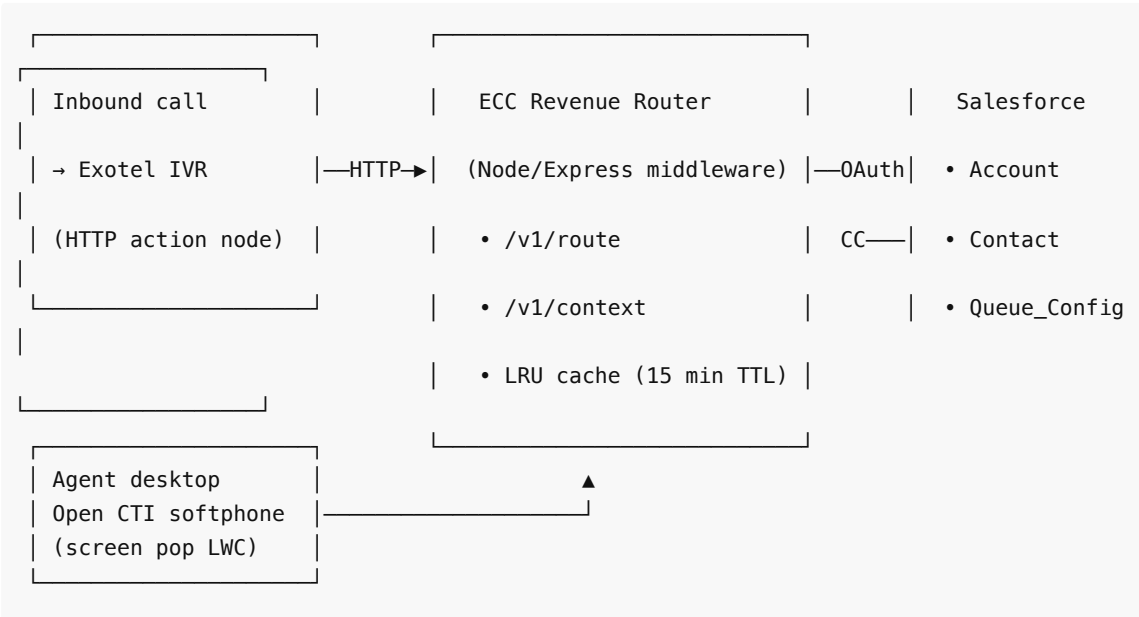
Audience: Professional Services, CX Ops, Support Leadership **Owner:** Shivanand (shivanand@exotel.com)
Branch: claude/upbeat-ride **Status:** Salesforce side built; awaiting PS engagement for telephony wiring.

1. Executive summary

Every inbound support call and every new Salesforce Case will carry the account's revenue tier as metadata, so routing, priority, and screen pop become tier-aware without manual agent lookup. Tiers are derived from existing Salesforce flags — no new data entry. The Salesforce + middleware side is fully built; PS needs to wire the Exotel IVR and Open CTI softphone to consume it.

Tier	Driver (on Account)	Target queue
HYPERCARE	Hypercare_active__c = 'Active' AND Hypercare_End_Date__c >= NOW()	Hypercare_Support_Queue
STRATEGIC	Top_Customer__c = TRUE OR Is_SAAS_Premium_Account__c = TRUE	Strategic_Support_Queue
PREMIUM	SAAS_Enterprise_Account__c = TRUE AND RAG__c IN ('GREEN', 'AMBER')	Premium_Support_Queue
STANDARD	everything else	FLR_Support_Queue
UNKNOWN	no Account link	FLR_Support_Queue (fallback)

2. Architecture



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New Case → Before-Save Flow → stamps Account_Revenue_Tier__c,
Account_RAG_at_Creation__c, Account_CSM_Name__c, Is_Top_Customer__c
→ Queue_Config__mdt drives assignment

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Two independent paths share the same tier model:

- **Call path:** IVR → Router → Salesforce phone lookup → tier → queue name → IVR transfers the call to the Salesforce-bound Exotel queue.
- **Case path:** Case insert → Before-Save Flow reads Account `Support_Tier__c` formula → stamps tier fields → standard Case assignment rules route to the tier queue.

The middleware is stateless apart from the in-process LRU cache, so it scales horizontally behind any HTTP load balancer.

3. Component inventory

Already built (this branch)

Layer	Artifact	Path
Account formulas	Support_Tier__c, Support_Tier_Badge__c	force-app/main/default/objects/Account/fields/
Case fields (x6)	tier, RAG snapshot, CSM, LTM revenue, Top Customer flag, badge	force-app/main/default/objects/Case/fields/
Case Flow	Case_RevenueTier_BeforeSave Before-Save	spec in 02-case-flow-specification.md
Queue config	Queue_Config__mdt + 4 records	force-app/main/default/objects/Queue_Config__mdt/, force-app/main/default/customMetadata/
Apex tests	10 tests incl. 200-record bulk	force-app/main/default/classes/Case_RevenueTier_FlowTest.cls
Connected App	ECC_Revenue_Router (OAuth 2.0 CC)	force-app/main/default/connectedApps/
Permission Set	ECC_Revenue_Router_Integration (read-only)	force-app/main/default/permissionsets/
Middleware	Node/Express service, Dockerized	middleware/
UAT matrix	53 test cases	06-test-and-rollout.md

Pending (PS or ops)

Layer	Item	Owner
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Telephony	IVR nodeflow HTTP action node + branch logic	PS
Telephony	4 Exotel queue destinations bound to SF queues	PS + CX Ops
Agent desktop	Open CTI softphone manifest update (banner + pop)	PS
Salesforce	4 SF Queues created with team membership	Support Leadership
Salesforce	Connected App Client Credentials Flow enabled + Run As user	Salesforce admin
Infra	Host middleware (container platform, DNS, TLS)	Internal SRE

4. Deployment plan — phased

Phase 0 — Pre-flight (before any deploy)

- ☐ Verify `Hypercare_active__c` picklist value is `Active` in the target sandbox. (Spec assumed value; one-line fix if different.)
- ☐ Run the phone-field audit SOQL from [01-field-definitions-and-phone-audit.md](#) and confirm ≥ 70 % phone-to-account match rate on last 30 days of calls. Below that, pause and clean up phone data first.
- ☐ Confirm the four target SF Queues exist (or create them) with the right agent membership.

Phase 1 — Salesforce deploy (UAT sandbox)

- ☐ `sf project deploy start -d force-app/main/default from claude/upbeat-ride .`
- ☐ `sf apex test run -n Case_RevenueTier_FlowTest -w 10 -c -- expect 10/10 pass, ≥ 75 % coverage on the Flow-adjacent code paths.`
- ☐ Manually create 5 Cases (one per tier scenario) and confirm field stamping in the UI.

Phase 2 — Middleware deploy (UAT)

- ☐ Open the deployed `ECC_Revenue_Router` Connected App → "Manage" → enable Client Credentials Flow → set Run-As user to the dedicated integration user with `ECC_Revenue_Router_Integration` assigned.
- ☐ Capture Consumer Key + Secret.
- ☐ `docker build -t ecc-revenue-router:uat middleware/`
- ☐ Deploy to UAT host, populate `.env` with Consumer Key/Secret + `ROUTER_SHARED_KEY` .
- ☐ `curl -H "X-Router-Key: ..." https://<uat-host>/v1/context?phone=%2B91... -- expect 200 with tier populated for a known account.`

Phase 3 — IVR + CTI wiring (PS-led)

- ☐ PS adds the HTTP action node to the target IVR nodeflow (design in [03-ivr-nodeflow-design.md](#)).
- ☐ PS binds the four Exotel queue destinations to the SF queues.
- ☐ PS updates the Open CTI softphone manifest per [04-screen-pop-and-layouts.md](#).
- ☐ End-to-end smoke: dial into each tier's test number, confirm queue transfer + banner render.

Phase 4 — UAT (1 week)

- ☐ Execute the 53-case UAT matrix in [06-test-and-rollout.md](#).
- ☐ Daily stand-up: review failed calls, false positives on hypercare, cache-staleness complaints.
- ☐ Sign-off: Support Leadership + CX Ops.

Phase 5 — Production rollout (staged)

- ☐ Deploy Salesforce metadata via change set or SFDX to prod.
- ☐ Deploy middleware to prod cluster (2 replicas min).
- ☐ Flip IVR nodeflow to prod — **single hotline first**, monitor 24 h.
- ☐ Expand to all support hotlines in a rolling rollout.

5. Dependency matrix

Blocks	Blocked by	Owner
Phase 1	Phase 0 pre-flight	Us
Phase 2	Phase 1, Salesforce admin enabling Client Credentials Flow	Us + SF admin
Phase 3	Phase 2, PS capability confirmation (auth scheme, latency, call_sid)	PS
Phase 4	Phase 3	Support + us
Phase 5	Phase 4 sign-off	Us + PS

6. Rollout & rollback

Rollout guardrails

- Start with **one hotline** in prod for 24 hours before expanding.
- Monitor: middleware p95 latency, Salesforce API error rate, cache hit rate, and first-response time per tier on new Cases.

Rollback — IVR

- Single flag in the IVR nodeflow: disable the HTTP action node's branch and fall back to the pre-existing default queue. Reversible in < 5 min.

Rollback — Case Flow

- Deactivate `Case_RevenueTier_BeforeSave` Flow in Setup. Existing Cases keep their stamped values; new Cases revert to default assignment.

Rollback — middleware

- Route the IVR back to the legacy path. The middleware is stateless; leaving it running or taking it down has zero Salesforce side-effects.

7. Success metrics (30-day review)

Metric	Baseline	Target
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Time-to-agent for HYPERCARE / STRATEGIC calls	(current)	–40 %
First-response time on HYPERCARE Cases	(current)	< 15 min
% of Cases with correctly stamped tier	0 %	> 99 %
Middleware uptime	N/A	> 99.9 %
Phone-to-account match rate	(from phone audit)	> 85 %

8. Open capability questions for PS

Full list in [05-ps-engagement-brief.md](#). Top three blockers for the UAT build:

1. **Auth scheme for the HTTP action node** — does Exotel support arbitrary header pass-through (X-Router-Key)? If not, what's the supported equivalent (token in query string, basic auth)?
2. **Call-transfer latency budget** — what's the acceptable upper bound for the HTTP action node's round-trip? We currently target < 300 ms p95; need confirmation the IVR can tolerate that without call quality issues.
3. **call_sid pass-through** — can the HTTP node include the call SID in the POST body so we can correlate router logs with Exotel CDRs?

9. Appendices (detailed docs)

- [README.md](#) — executive summary
- [01-field-definitions-and-phone-audit.md](#) — SF fields, phone audit, normalization
- [02-case-flow-specification.md](#) — Before-Save Flow
- [03-ivr-nodeflow-design.md](#) — IVR + middleware API
- [04-screen-pop-and-layouts.md](#) — CTI softphone + page layouts
- [05-ps-engagement-brief.md](#) — full 10 capability questions
- [06-test-and-rollout.md](#) — UAT matrix + runbook
- [middleware/README.md](#) — middleware ops manual